Simple Pendulum Length Calculation

Problem Statement

Practice Ex. 5.13: Calculate the length of a simple pendulum to make 120 complete oscillations per minute.

Solution

Step 1: Determine the Time Period

The time period T of a pendulum is given by:

$$T = \frac{1}{f}$$

where f is the frequency in Hz. Given that the pendulum makes 120 oscillations per minute:

$$f = \frac{120}{60} = 2 \text{ Hz}$$

Thus, the time period is:

$$T = \frac{1}{2} = 0.5 \text{ s}$$

Step 2: Use the Formula for a Simple Pendulum

The formula for the time period of a simple pendulum is:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where:

- l is the length of the pendulum,
- $g = 9.81$ m/s^2 (acceleration due to gravity).

Rearranging for l:

$$l = \frac{T^2 g}{4\pi^2}$$

Substituting the values:

$$l = \frac{(0.5)^2 \times 9.81}{4\pi^2}$$

Calculating:

$$l = \frac{0.25 \times 9.81}{39.478}$$

$$l = \frac{2.4525}{39.478} \approx 0.0621 \text{ m}$$

Conclusion

The required length of the pendulum is **0.062 m (6.2 cm)**.